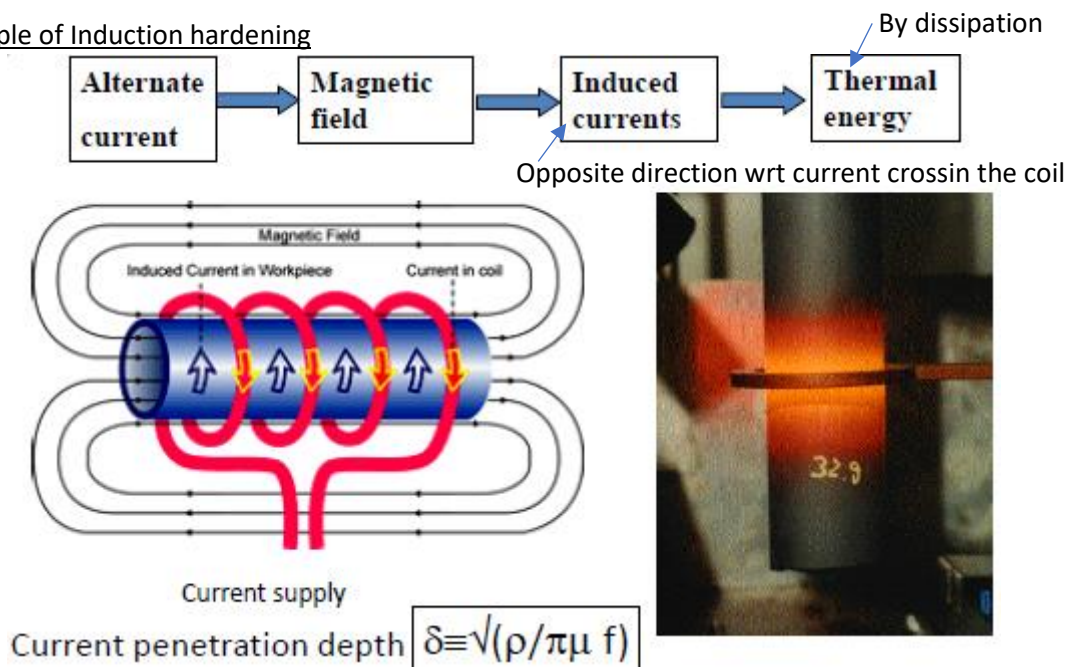


Heat Treatments for post-carburized and post-nitrided steel parts

Carburized	Nitrided
Normalizing +	Normalizing +
work hardening annealing	Work hardening annealing
Machining roughing (medium thickness)	Machining roughing (medium thickness)
Carburizing	Tempering
Quenching + tempering at 150 °C	Finish Nitriding
Finish by grinding	Finish by grinding

Basic principle of Induction hardening

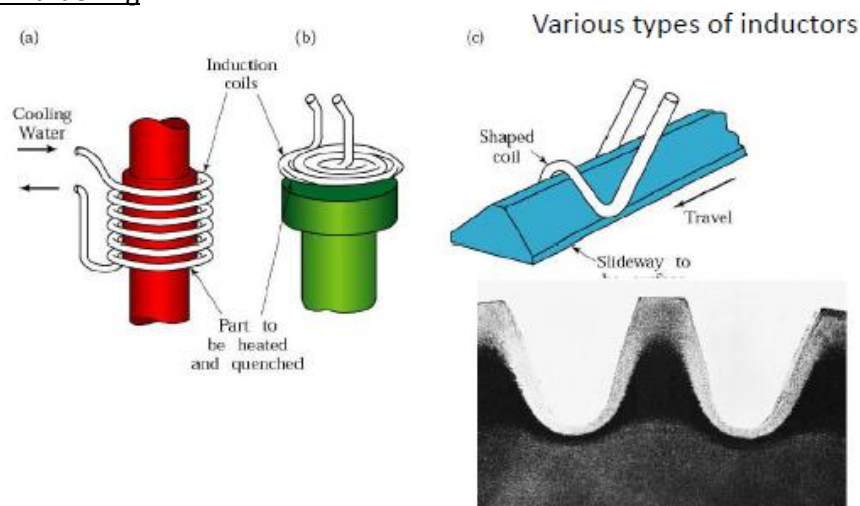


It doesn't exploit chemical addition but just heating the region to make it harder.

Heating by induction: insured by a coil crossed by current. We put our part inside the coil.

With frequency proportional to the resistivity of the material and inversely proportional to relative magnetic permeability, we have that the higher is the frequency, the more superficial is the heating of the part.

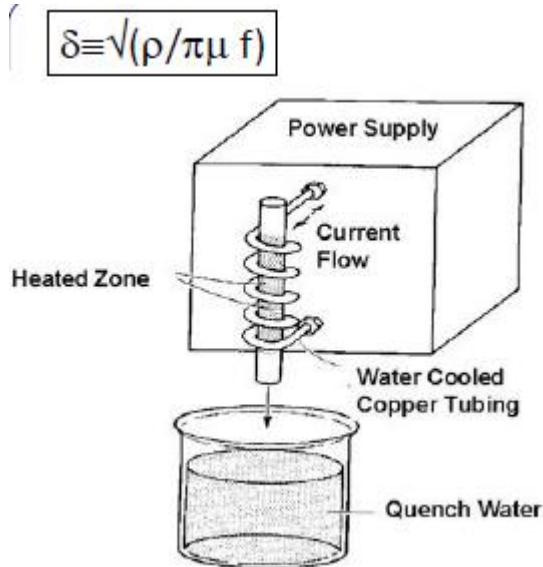
Induction Thermal Hardening



By changing the shape of the coil, we can have more parts subjected to this process.

The gear becomes harder after induction hardening because it's a normalized material. Annealed materials are not suitable for this because they will provide ferrite and pearlite colonies microstructure → **leopardizing effect** → hardening is not uniform.

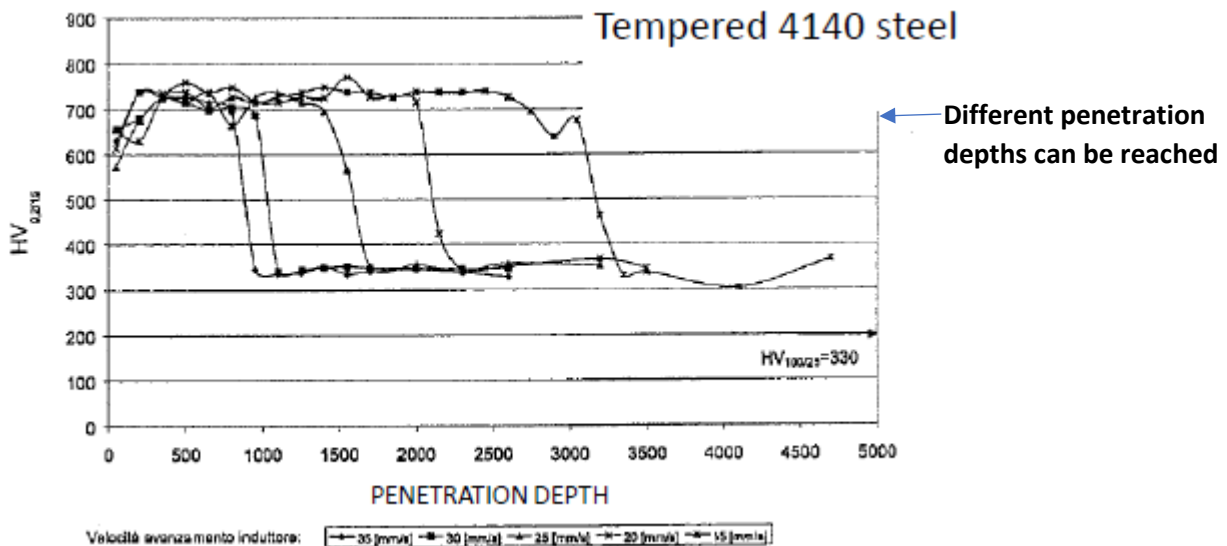
Induction hardening (single-shot)



Induction hardened parts

After the part is heated by induction, it is left to fall in the quenched water → **hardening**.

Induction heat treatment: hardness profiles along the thickness



Hardness by induction is not very high → much lower than Nitriding and a little bit higher than carbonizing.

Heat vs. Chemical Surface Treatments

PROCESS	STEELS	COMPONENTS	CONDITIONS
INDUCTION 3 kHz	0.38wt. % C	Axle	Water Quenched. Tempered at 210°C
FLAME HARDENING	0.67wt. % C	Wheel	Water Quenched. Tempered at 490°C
CARBURINZING.	0.2wt. % C-1.8wt. % Ni -0.5wt. % Cr-0.2wt. % Mo	Gear	Oil Quenched. Tempered at 200°C
Laser 15 kW	0.43wt. % C-1.5wt. Mn	Gear	Self Quenched
NITRIDING	0.2wt. % C-1wt. % Cr -0.1wt. % V-0.08wt. % Al	Gear	Process Carried out at 570°C

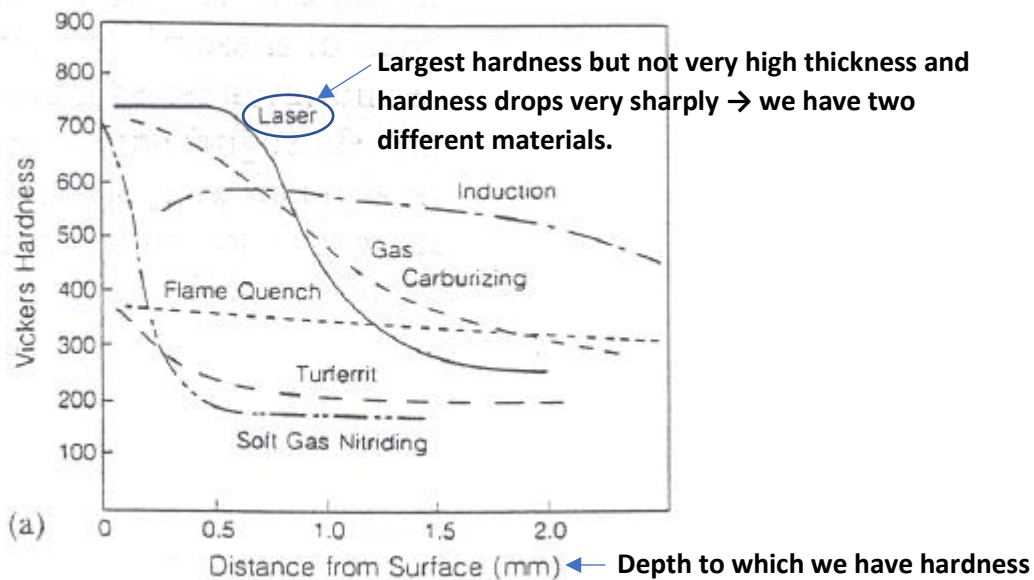
These surface heat treatments, beside originating a high surface hardness, will determine an increase in fatigue resistance as all surface heat treatments will confer a compressive stress state on the surface.

Fatigue: one of the most severe loadings of mechanical components.

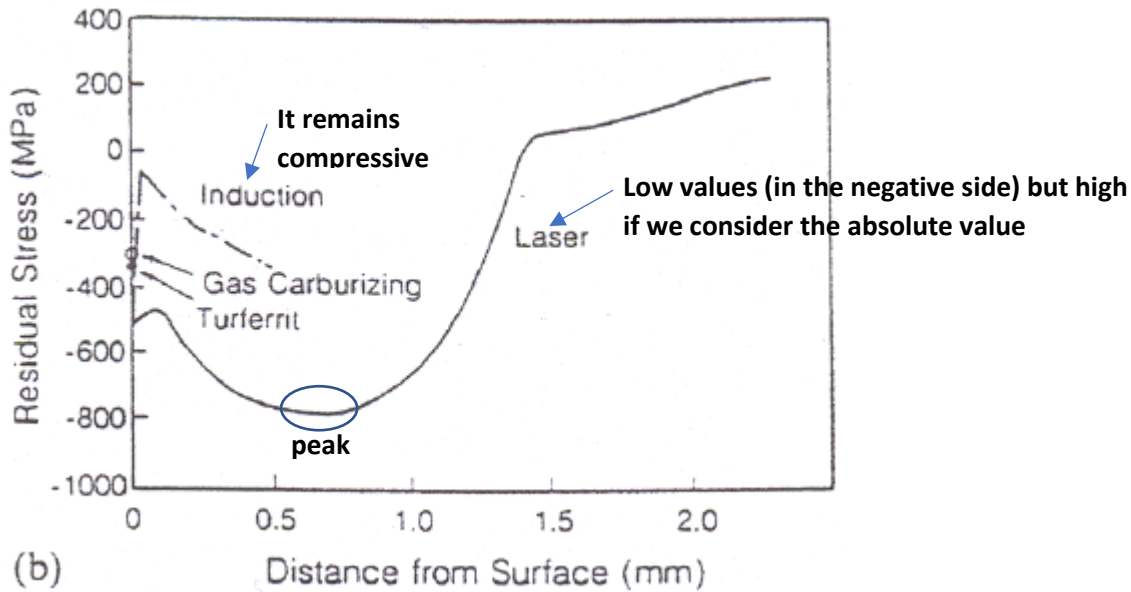
Residual stresses which resolve from heat treatments, give tensile stresses → if not performed well, they result in cracks. Due to the different situation of the surface compared to the bulk, the situation of the stress state here is reversed → on the surface, we have compressive stress state (an improvement from the fatigue point of view and from the crack propagation).

Flame hardening: old process → flame induces C inside surface.

Heat vs. Chemical Surface Treatments



Surface Heat vs. Chemical Surface Treatments

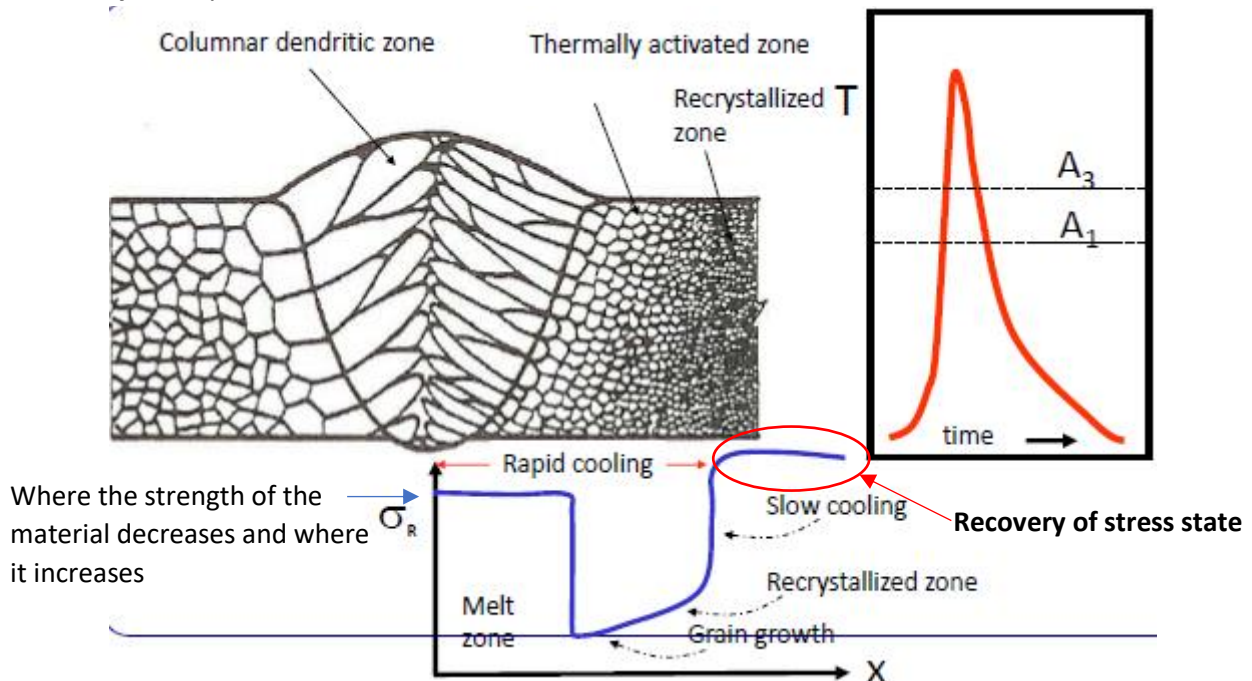


Despite the penetration distance of less than one mm of laser, we have inside the material a compressive state till it reaches 1,5% and then we have tensile state → compressive stress on surface and tensile in bulk. The smaller the penetration distance, the larger the stress state (like induction hardening).

Weldability of Steels

Welding: Joining together two metal parts by heating the surface to the point of melting. After the material has been melt, it will solidify.

Weldability: Ability of a material to be welded.



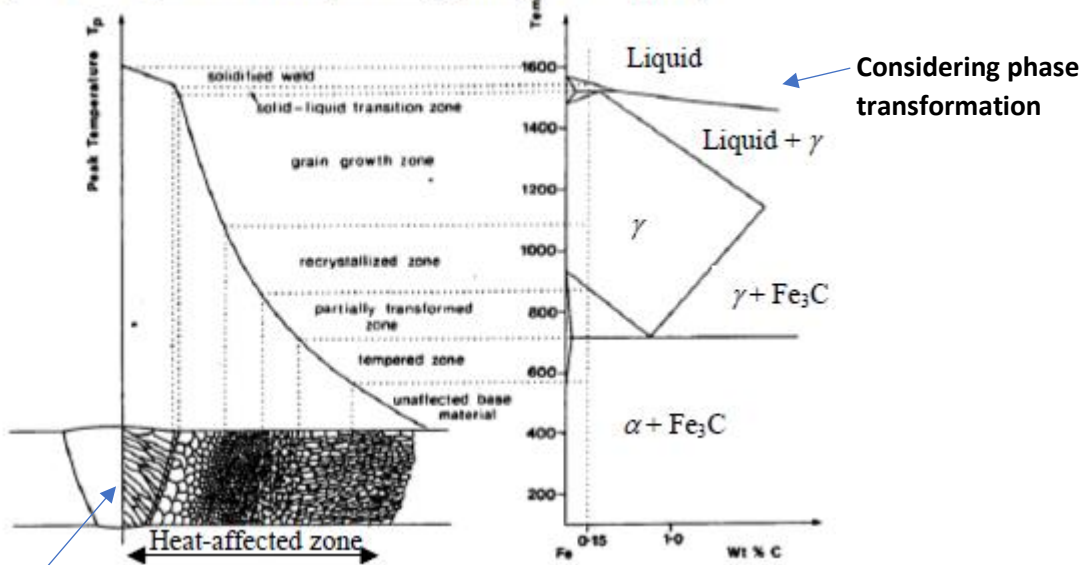
Problems:

- **Light weight construction** → we want to make the material thinner (more difficult for welding, the material melts easily)
- **Dissimilar material welding** → parts are made of different materials (different melting points)

Weldability of steels: HAZ (Heat Affected Zone)

Welding of hypoeutectoid steel (0.15 wt% C) vs Fe-Fe₃C diagram

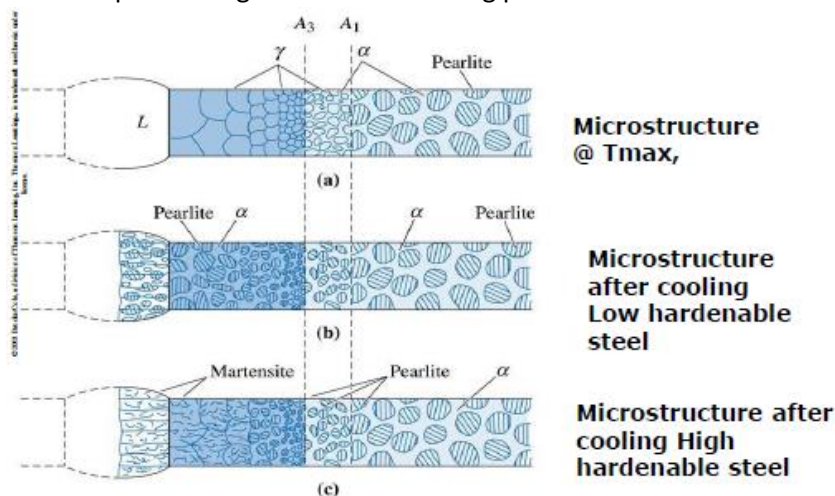
(From K. Easterling, *Introduction to the Physical Metallurgy of Welding*, Butterworths, (1983))



Once this region has been recrystallized, the base material which is far from the melting region will undergo a kind of heat treatment → if the material has been annealed or normalized, the region becomes tempered; after that, we have an unaffected base material.

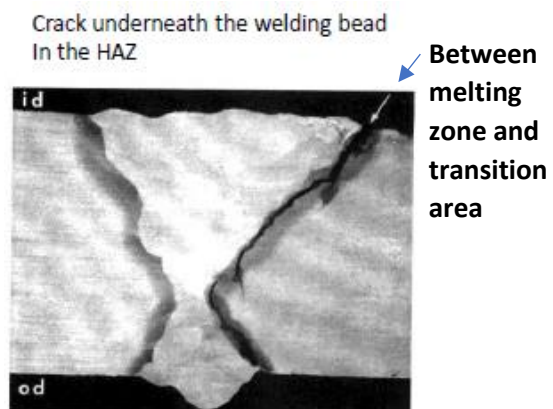
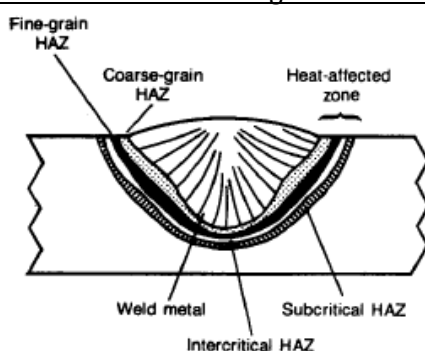
Formation of HAZ

How to identify the most important region from the melting pool to the base material.



The increase in temperature changes the microstructure; despite the elongated crystals in the melting zone, the cracks will occur in the transition area.

Microstructure in the welding zone: low carbon steels



In order a steel to be weldable, has to be low Hardenable; it must contain low content of alloying elements; a steel having up to 0.4%C can be weldable although caution is needed in steels with higher carbon content.

Chemical Surface Treatment: Summary

Process	Nitriding	Nitro-carb.	Boriding	Carburizing	Process QUAD
Superficial compound	Fe-N	Fe-(C,N)	Fe-B	Fe-C	CrCN,VCN, TiCN
T sup. °C	500-550	550-650	600-1000	700-1050	400-700
T core °C	500-550	550-650	600-1000	700-1050	400-700
Proc.time(h)	1-70	1-6	1-4	1-46	1-8
Thickness μm	50-100	50-200	50-500	100-7000	5-20
hardnessHV	700-1000	700-1000	1000-1800	750-950	1200-2500
Distorsion	min	min	high	high	min
Initial H.T.	yes	yes	after	after	yes
Post-trattam.	seldom	seldom	seldom	needed	seldom
Unif. Spess.	good	good	good	good	good

Wear resistant ceramic coatings

Process	Cr hard	Thermal CVD	Plasma CVD	PVD	Flame coating	Stellite
Surface layer	Cr	TiC, TiCN, TiN	TiC, TiN	TiN, CrN	Ni, Cr, B, Si	Ni, Cr, B, Si
Method	electrolysis	gas	Arc in vacuum	N in vacuum	Powder melting in plasma	melting (arc)
Surface T°C	50-80	800-1100	300-600	300-600	1000-1100	$\approx T_{fus}$
Core	50-80	800-1100	300-600	300-600	500-800	300-900
Time (h)	1-6	4-8	1-3	2-4	dep. n size	dep. on size
thickness	20-50	3-15	1-6	1-6	500-2000	2000-5000
HV hardness	700-800	1500-2000	1500-2000	2000-4000	600-800	300-900
Distorsion	low	high	low	low	high	high
Substrate	before	after	before	prima	before or after	Before or after
Post-treatment.	seldom	seldom	seldom	seldom	needed	needed
Uniformity	low	good	poor	poor	Very poor	Very poor
Substrate.	Met./non	Metals	Metals	Metals	Metal/nonMet	Metals

Flame coating vaporizes any sort of material.

Instead of enriching the surface in order to increase the hardness, we may deposit something from the environment and we may use plasma, for instance, to heat up the surface (plasma ensures much higher T).

Lecture 14

Aluminum Alloys and Magnesium Alloys

General Properties of Aluminum

Aluminum is widely used in the industry, mainly to produce castings. Pure Al is expensive but thanks to recycling, Al has become more affordable. Secondary Al should not have high content of Fe because it will cause problems in castings such as brittleness.

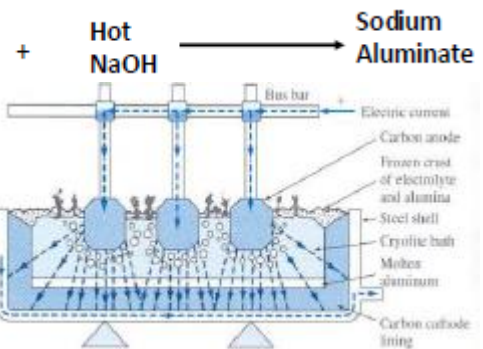
Al is characterized by:

- Low density, corrosion resistance (uniform protection of surface)
- **High alloy strength** (about 690 MPa) but low toughness
- Nontoxic and good electrical properties.

Production:

- Aluminum hydroxide $\text{Al}(\text{OH})_3$ is **precipitated** from aluminum solution.
- Aluminum hydroxide is thickened and calcined to Al_2O_3 which is dissolved into cryolite (a mineral) and **electrolyzed**.
- Metallic aluminum sinks to bottom (since it's heavier) and it tapped out.

Aluminum
Ore (Bauxite)



For every alloy, we usually have to face two issues:

- Classification and designation
- How alloys can be produced in order to be exploited → suitable processes to shape the alloy.

Alloys are produced in two ways:

- **Castings** (melting → solidification)
- **Wrought** (beaten out or shaped by hammering, plastic deformation, rolling, forging, ...)

Wrought Aluminum Alloys

- **Primary Fabrication:** Usually direct chill method.
- **Scalping:** ½ inch metal is removed from hot rolled surface for goof finishing.
- Ingots are homogenized and rolled.
- **Classification:** According to major alloying elements. We have **four digits**: the first digit indicates the major group of alloying elements; the **second digit** indicates the impurity limits; the **last two digits** identify the aluminum alloy (grade of alloy).

Aluminum, 99.00% minimum and greater	1xxx
Aluminum alloys grouped by major alloying elements:	
Copper	2xxx
Manganese	3xxx
Silicon	4xxx
Magnesium	5xxx
Magnesium and silicon	6xxx
Zinc	7xxx
Other element	8xxx
Unused series	9xxx

Al-Li is very light, and it's used in aerospace application.

Temper Designations

- Temper designations are designated by **hyphen**.
- **Example**: 2024-T6

F – as fabricated
O – Annealed
H – Strain hardened.
T – Heat treated to produce stable temper

H1 – Strain hardened alloy.
H2 – Strain hardened and partially annealed.
H3 – Strain hardened an annealed

T1 – Naturally aged
T3 – Solution heat treated.
T4 – Solution heat treated and naturally aged.
T5 – Cooled and artificially aged.
T6 – Solution heat treated and artificially aged.
T7 – Solution heat treated and stabilized.
T8 – Solution heat treated, cold worked and then artificially aged.

Non Heat Treatable Aluminum Alloys

- **1xxx alloys**: 99%Al + Fe + Si + 0.12%Cu
Tensile strength = 90 MPa (lowest strength → soft → more suitable to make passivation of surface, used in kitchen utensils as it doesn't react with food especially when hot)
Used for sheet metals.
- **3xxx alloys**: Manganese is principle alloying element.
AA3003 = Al 1100 + 1.25%Mn
Tensile strength = 110 MPa (strength increased)
General purpose alloy
- **5xxx alloys**: Al + up to 5% Mg
AA5052 = Al + 2.5%Mg + 0.2%Cr
Tensile strength = 193 MPa
Used as sheet in automotive, bus, truck and marine

5xxx is more used.

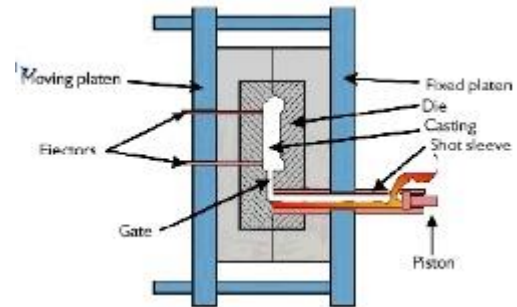
Heat Treatable Aluminum Alloys

Higher amount of Mg → alloy is heat treatable.

- **2xxx alloys**: Al + Cu + Mg
AA2024 = Al + 4.5%Cu + 1.5%Mg + 0.6%Mn
Strength = 442 MPa
Used for aircraft structures
- **6xxx alloys**: Al + Mg + Si
AA6061 = Al + 1%Mg + 0.6%Si + 0.3%Cu + 0.2%Cr
Strength = 290 MPa
Used for general purpose structure, in construction to make extruded parts (windows, doors, ...)
- **7xxx alloys**: Al + Zn + Mg + Cu (larger amount of alloying elements)
AA7075 = Al + 5.6%Zn + 2.5%Mg + 1.6%Cu + 0.25%Cr
Strength = 504 MPa
Used for aircraft structures.
We can use aging to make the alloy harder.

Aluminum Casting

- **Sand Casting:** Simple and used for small quantities and complex jobs. The mold is broken down after a single use. It is characterized by a slow solidification process.
- **Permanent mold casting:** Molten metal is poured into permanent metal mold. Molds are made of metal.
 - Finer grain structure and strength due to fast cooling.
 - Less shrinkage and porosity.
 - More shrinkage and simple parts only.
- **Die casting:** Molten metal forced into molds high P. The metal mold is made by two parts, one of which is sliding. At the conjunction line, we can have leakage of molten metal that is reused again.
 - Almost finished parts, automatic.
 - Good tolerance and surface finish.
 - Low porosity and low shrinkage.



Die casting is the most used in automotive.

How to strengthen Al castings: to refine eutectic microstructure

by adding Strontium and refine grains of casting by adding Titanium → improvement of strength and toughness of castings.

Aluminum Casting Alloys

- Composition of casting alloys differs greatly from wrought alloys
- Casting and mechanical properties are of primary interest.
- Denoted as 4 digits with a period between last two digits.

Aluminum, 99.00% minimum and greater	1xx.x
Aluminum alloys grouped by major alloying elements:	
Copper	2xx.x
Silicon, with added copper and/or magnesium	3xx.x
Silicon	4xx.x
Magnesium	5xx.x
Zinc	7xx.x
Tin	8xx.x
Other element	9xx.x
Unused series	6xx.x

Further Strengthening of Aluminum Alloys (Aging)

- **Precipitation Strengthening:** Creates fine dispersion of precipitated particles in the metal and hinder dislocation movement.
- Basic steps:
 - **Solution heat treatment:** Alloy sample heated to a temperature between solvus and solidus and soaked at that temperature.
 - **Quenching:** Sample then quenched to room temperature in water.
 - **Aging:** Solutionized and quenched sample is then aged to form finely dispersed particles.

Decomposition Products Created by Aging

- Super saturated solid solution is in METASTABLE condition.
- Alloy tends to seek a lower energy state by decomposing into metastable or equilibrium phase.
- Supersaturated solid solution as **highest energy** state.
- Equilibrium precipitate has **lowest energy** state.

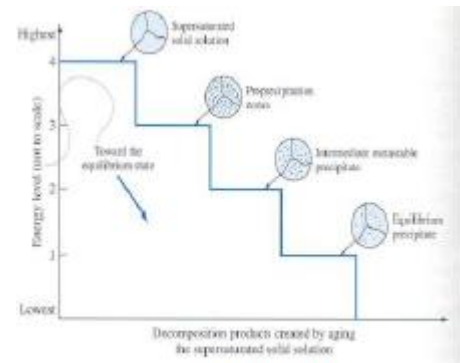


Figure 9.42

Effects of Aging on Strength

- **Aging curve**: Plot of strength or hardness versus aging time.
- As aging time increases alloy becomes **stronger** harder and **less ductile**.
- **Overaging** decreases strength and hardness.

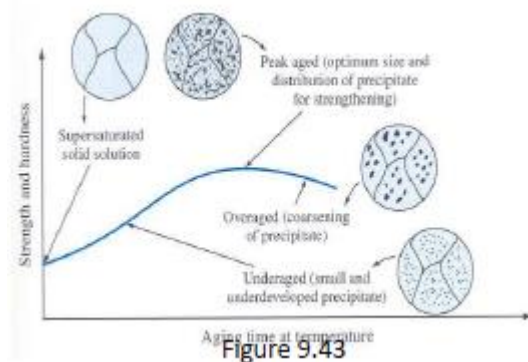


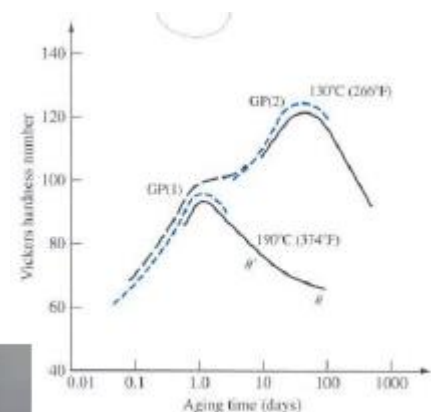
Figure 9.43

Example – Al 4% Cu Alloy

- Al 4% Cu is **solutionized** at about 515 0 C
- Alloy is rapidly cooled in water.
- Alloy is artificially aged in 130 190 0 C
- Structures formed:
 - **GP1 Zone**: At lower aging temperature, copper atom is segregated in supersaturated solid solution.
 - **GP2 Zone**: Tetragonal structure, 10 100 nm diameter.
 - **θ' Phase**: Nucleates heterogeneously on dislocation.
 - **θ Phase**: Equilibrium phase, incoherent (CuAl_2)

Correlation of Structure and Hardness

- GP1 and GP2 Zones increases hardness by stopping dislocation movement.
- At 130 0 C when θ' forms, hardness is maximum.
- After θ' forms, GP2 zones are dissolved and θ' gets coarsened reducing hardness



Magnesium Alloys



Restricted to only some parts.

Magnesium



- Magnesium is the **lightest** of the commonly used metals (density is 1.7 g/cm³). Its **versatility** (by both casting and rolling) and **abundance** make it one of the most promising materials for **efficient lightweight applications**.
- Magnesium alloys are also **impact and dent resistant**, and they have the capacity for **damping and low inertia**, which makes them effective in high speed applications. **Poor plastic deformation, poor cold rolling due to HCP structure; usually hot worked**.
- Like the majority of common metals, magnesium is **too soft to be used as a structural metal**. Magnesium is most **commonly alloyed with aluminum**.
- Due to **easy machinability** and **ease of casting**, magnesium is used for many purposes such as **auto parts**, **sporting goods**, **power tools**, **aerospace equipment**, fixtures, and material handling equipment **electronic casing**. Automotive applications include gearboxes, valve covers, **wheels**, clutch housings, and brake pedal brackets. Wrought alloys are available in rod, bar, sheet, plate, forgings and extrusions.

Alloy designation

- Two alloy classes:
 - **wrought alloys** (sheet, plate, extrusion)
 - **casting alloys** (casting).
 - Designated by:
 - First 2 capital letters: two major alloying elements
 - The 2 or 3 digits: **wt%** of alloying elements.
- AZ93C**: A=Al; Z = Zn; 9 = 9 wt.% Al; 3 = 3wt.% Zn; C= alloy modifications
Al and Zn are added to **increase strength**.
Alloying with rare earth elements (cerium) produces **rigid boundary network**.
Typical tensile strength: 179 310 MPa and elongation: 2 to 11%

Processing designation

- **F**: as produced (without any heat treatment)
- **O**: Annealed
- **H10, H11**: Slightly strain hardened
- **H23, H24, H26**: Strain hardened and partially annealed
- **T4**: Solution heat treated
- **T5**: Artificially aged only
- **T6**: Solution heat treated and artificially aged
- **T8**: Solution heat treated, cold worked, and artificially aged

Typical alloys

AZ91D – Al (9%), Zn (1%), D – fourth modification of the alloy
AM50A – Al (5%), Mn (<1%), A – first modification of the alloy.
AM60B – Al (6%), Mn (<1%), B – second modification of the alloy.

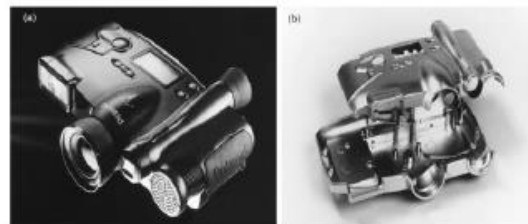


Figure extra (a) The Polaroid PDC-2000 digital camera with a AZ91D die-cast, high purity magnesium case. (b) Two-piece Polaroid camera case made by the hot-chamber die casting process. Source: Courtesy of Polaroid Corporation and Chicago White Metal Casting, Inc.

Lecture 15

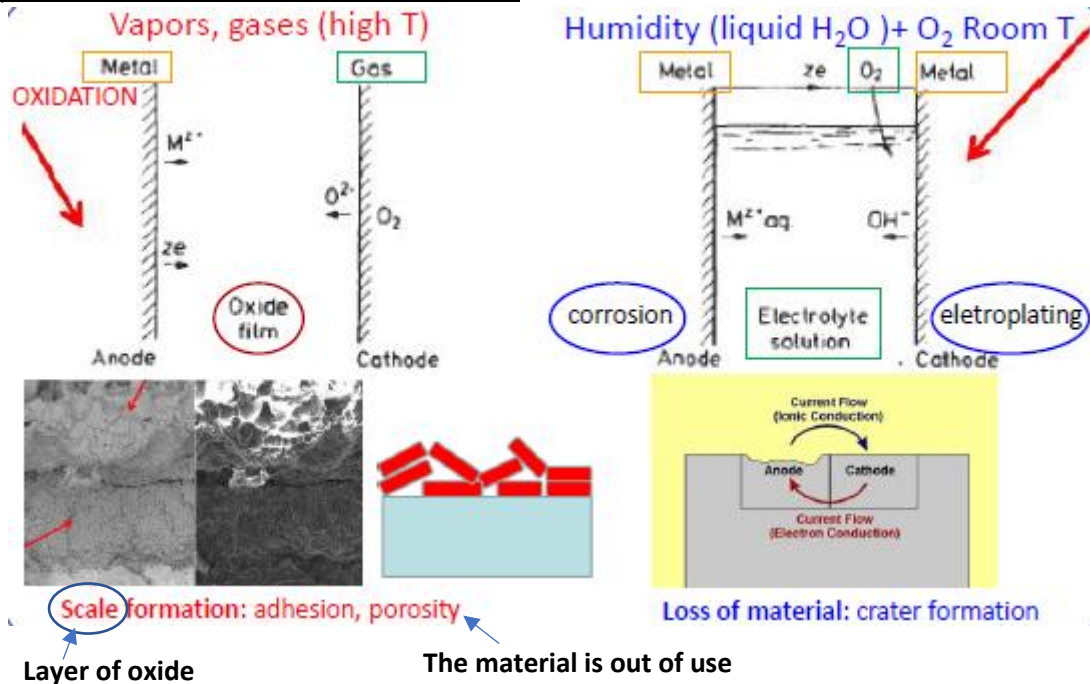
Corrosion and Stainless Steels

Corrosion phenomena

It occurs in steels as a result of facing atmosphere, mainly humidity.



Main types of corrosion: DRY and WET CORROSION



In wet corrosion, water works as an electrolyte, it contains ions and can be decomposed and move like in an electric circuit $\rightarrow$ ions will conduct the current into the electrolyte.

Electroplating $\rightarrow$ a layer of metal is deposited on metallic or non-metallic electrodes.

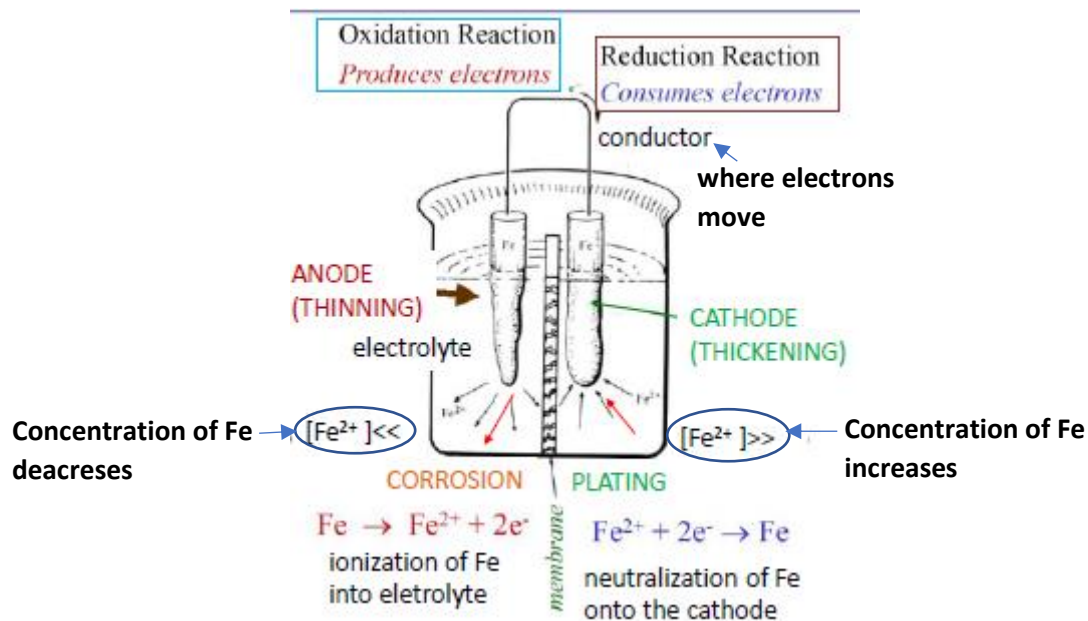
Corrosion in Iron and steels

The **anode** gives up electrons to the circuit and corrodes.

The **cathode** receives electrons from the circuit by means of a chemical or cathode reaction. By-products may form.

Anode, cathode, conductor and electrolyte must be electrically connected to promote corrosion.

In general, the moisture near the metallic parts, in the presence of the oxygen in the environment, causes corrosion.



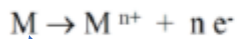
The circuit is closed by the presence of water.

Oxidation-reduction processes take place and metal atoms lose electrons (become ions) and go into solution. Often occurs in aqueous medium where moisture can provide electrical circuit to form an **electrochemical cell**.

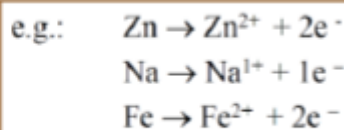
Anodic reaction

Oxidation reaction – Corrosion

Metals $\rightarrow$ Ions



Neutral



Metallic portions of the anode electrode precipitate as ions into the electrolyte solution (whereas the released electrons flow through the conductor towards the cathode).

Cathodic reaction

Reduction reaction

This type of reaction depends on the local condition at the cathode:

Electrons conducted by the wire

- ✓ If H^+ ions are available then: $2H^+ + 2e^- \rightarrow H_2$
- ✓ If acid solution with dissolved oxygen: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
- ✓ If basic/neutral with dissolved oxygen: $O_2 + 2H_2O + 4e^- \rightarrow 4(OH^-)$

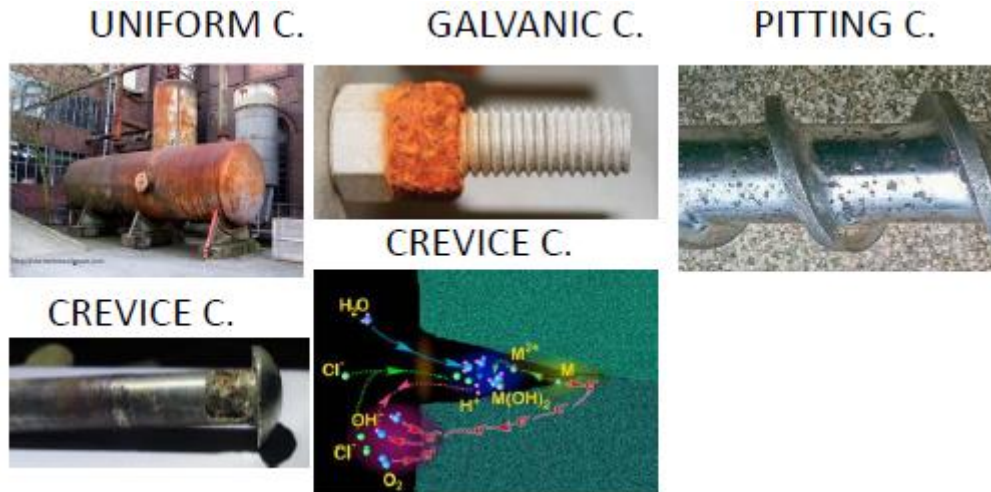
If some metal ions are present, they may be reduced by the electrons:



Cathode will be enriched with metal

Can get more than one of these occurring at the same time.

Classification of corrosion phenomena



Uniform Corrosion: it's the most frequent but it's not the main cause of failure of components because it's slow so we can make preventions.

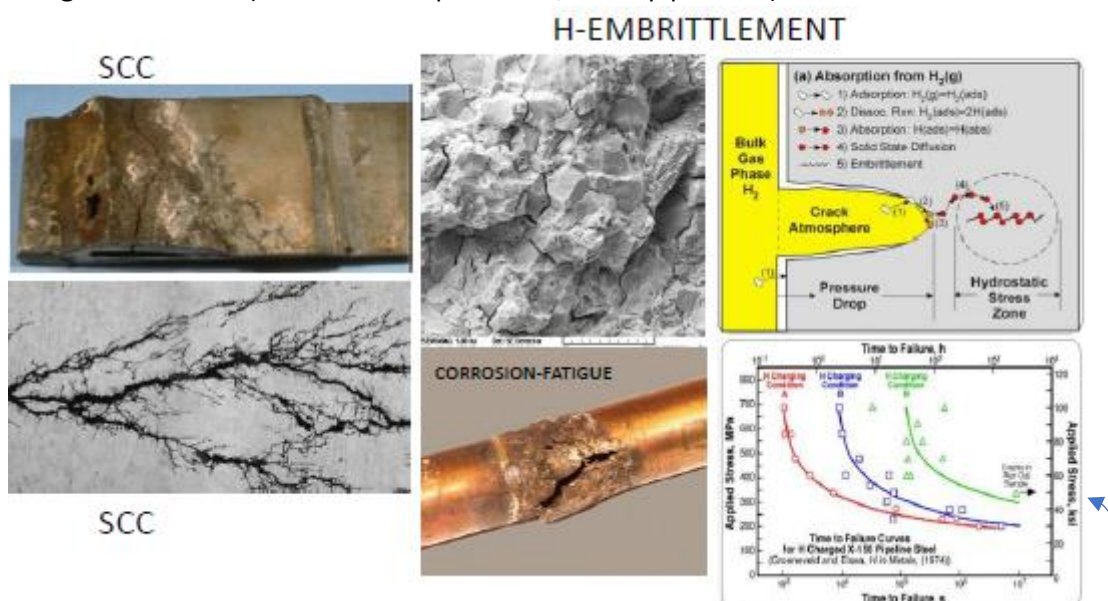
Galvanic Corrosion: occurs in couplings of metals.

Pitting corrosion: it may start from defects. It's highly localized so it's dangerous.

Crevice Corrosion: it occurs when designers make mistakes, a kind of interstices between couplings (water enters inside).

Synergetic corrosion phenomena (more dangerous)

- **Stress Corrosion Cracking** – enhanced by external loads
- **Hydrogen Embrittlement** – enhanced by the presence of H (from external sources or formed in site during corrosion, welding, etc.). Hydrogen on its own is very dangerous (mono-atomic, small atomic size, ...) and it diffuses into the solid especially when temperature increases.
- **Corrosion Fatigue** – attack is affected by mechanical factors
- **Selective Leaching** – one element of the alloy is removed by corrosion.
- **Fretting Corrosion** – low frequency sliding + corrosion
- **Erosion Corrosion (wear)** – attack is modified by flow effects.
- **Microbiological Corrosion** (ex. biomedical prosthesis, water pipes. Etc.)



Different concentrations of H → time to failure will make applied stress lower and lower because H will reduce stress applied to the part.

Uniform corrosion (most common)

- **Relevance:** responsible for the greatest destruction of structural materials, but technically **not a major cause for concern**
- **Controlling factor:** the relative extent of the anodic and cathodic areas
- **Mechanism:** **corrosion cells are very small, numerous, and randomly distributed over the surface of the metal, such that the effect is more or less of uniform attack on the surface**
- **Rate of corrosion:** proportional to the flow of electron current which may depend on a number of factors, but **the important criterion** is the balance between the rates of anodic and cathodic reactions required to maintain electro neutrality whichever reaction occurs least readily will control the corrosion rate).

Galvanic or bimetallic corrosion

- **Two dissimilar metals in relative contact and immersed in a conducting solution** usually develop **different corrosion potentials**, also in ne alloy fabricated with lamellar structure (like pearlite, which will be consumed because α and cementite have different electric potential $\rightarrow$ current $\rightarrow$ corrosion rate).
- Such potential difference is the driving force for corrosion:
 - the less noble of the two metals corrodes more rapidly than the other
- A standard **electrode potential table**, based on standard galvanic series data for the metals with respect to H potential, is commonly used for electrochemical ranking of the metals:
 - the **table** shows which alloy of the selected galvanic couple tends to be either anodic or cathodic
 - however, **the tables give no indication on the rate of corrosion**
- **Surface passivity** of metals (the state of inactivity of a metal made unreactive by a thin inert surface layer of oxide) markedly affects the corrosion rate, by decreasing it, but this effect is not taken into account in the table.

Standard electrode potentials (conditions: 300K, molar solution)

Oxidation reaction for solution of the metal	Normal hydrogen scale (volts)
$\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$	- 2.36
$\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$	- 1.66
$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$	- 0.76
$\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^-$	- 0.74
$\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$	- 0.44
$\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$	- 0.25
$\text{Sn} \rightarrow \text{Sn}^{2+} + 2\text{e}^-$	- 0.14
$\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$	- 0.13
$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	0.00
$\text{Sn}^{2+} \rightarrow \text{Sn}^{4+} + 2\text{e}^-$	+ 0.15
$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$	+ 0.34
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4(\text{OH})^-$	+ 0.40
$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$	+ 0.77
$\text{Ag} \rightarrow \text{Ag}^+ + \text{e}^-$	+ 0.80
$2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$	+ 1.23
$\text{Au} \rightarrow \text{Au}^{3+} + 3\text{e}^-$	+ 1.42

← ranking of metals according to their potential corrosivity.

Mg is the most prone to corrosion.

Al is appealing in automotive applications because we don't need to paint it, it protects itself because it makes passivating films. However, from an electronic point of view (the table), it is highly reactive but fortunately it will oxidize and make a self-protecting coating. Since the coating is uniform, dense and cohesive with the substrate $\rightarrow$ Al will not be subjected to corrosion.

Cr as well is self-protecting.

Factors affecting Galvanic Corrosion

- **Environment / Separation effect**
 - galvanic effects are reduced in proportion to the **distance from the cathodic region** and to the **resistivity** of the environment
 - the attack at the anodic regions is sensibly reduced in **poorly conducting solutions**, where the electric resistance of the corrosion cell plays a major role in controlling the corrosion current

- **Ratio of the Surface areas effect (same surface area → corrosion)**

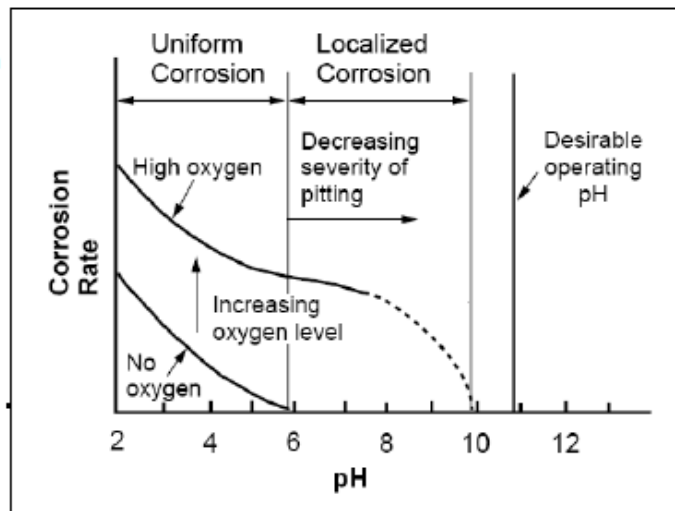
- is the key factor used to prevent galvanic corrosion by design defect
- if a large cathodic area is near to a small anodic area (with surface ratios of hundreds or even thousands), then the rate of attack of the anodic material may become extremely high indeed.
- Conversely, large surface anodes coupled to small surface cathodes may not undergo a sluggish corrosion.

- **Temperature of the environment**

- **PH of the environment**

Factors controlling corrosion rate other than electrode potential

- % O in solution
- PH
- Temperature



Localized Corrosion or Crevice Corrosion

-Favored by the presence of crevices, either as a result of (wrong) **design** or by formation of **deposits**, or the like

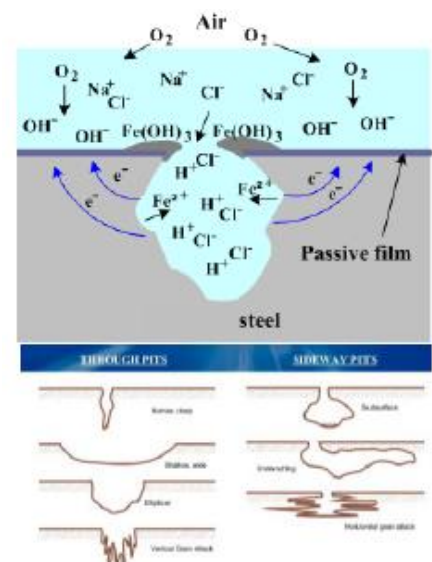
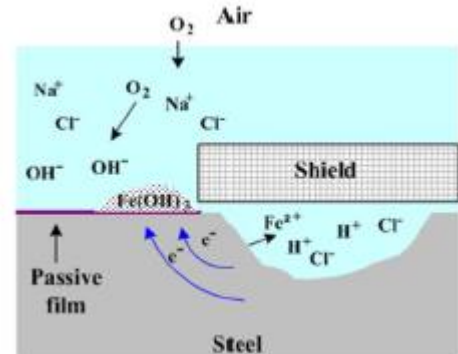
- Such obstructed **regions** in which oxygen, the usual cathodic reactant, cannot be replenished.

-Whereas both anodic and cathodic reactions once occurred uniformly over the surface, after a short period **the only reaction remaining in the crevice may be the anodic reaction**, balanced by **cathodic action outside the crevice**.

-It is **most serious in active/passive materials** where a sequence of events may follow leading to significant and irreversible changes in the environment within the crevice.

Small anode and large cathode → high corrosion speed.

We can't see whether there is corrosion or not under the shield (like tape) → problem.



Localized Corrosion: Pitting Corrosion

- **Extremely dangerous** because it is localized whereas **the overall metal loss is negligible**.

- It occurs predominantly as a result of the **action of chloride ions**, although bromides, hypo-chlorites and a few other species are **occasionally responsible**, and it is particularly insidious because it frequently occurs in **environments which are close to provide**

complete protection for the alloy except for occasional fluctuations in conditions.

- The phenomenon can cause **major changes of the environment within the pit**, when an **autocatalytic increase in corrosion rate** can occur.

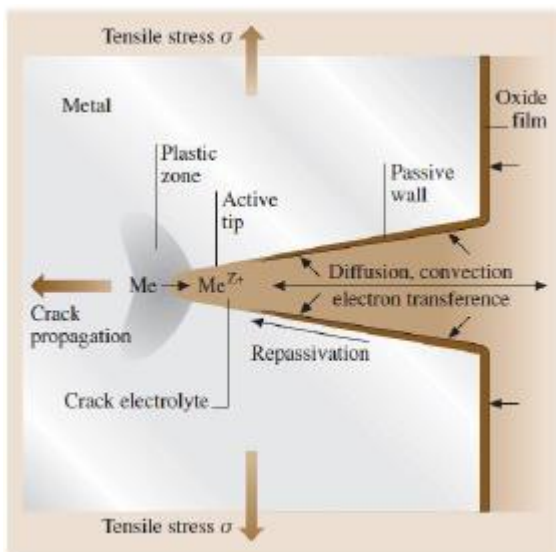
Pitting Corrosion is dangerous because the material has some weaknesses which are sources of heterogeneity, leading to corrosion.

Horizontal corrosion → part will crack very dramatically.

H-embrittlement

- Hydrogen embrittlement is a mode of **low ductility fracture** which may be induced **in ferritic and martensitic steels at high strength levels** and in **titanium alloys** by the introduction of hydrogen (it enhances crack propagation).
- Although it is relatively simple to demonstrate the distinction between hydrogen embrittlement and true stress corrosion in suitable laboratory tests it is by no means easy to separate the mechanisms in practical failure situations with high strength materials.
- Thus, for example much of the stress corrosion cracking reported in alloy steels in **hydrogen sulfide** or **cyanide** environments is undoubtedly **due to the effects of hydrogen rather than to anodic processes**.
- The hydrogen may arise from a **variety of sources** including electroplating and welding, but the problem is of greater concern when the hydrogen is generated by corrosion.
- Not only is this continuously available while the alloy is exposed, but for sufficiently susceptible alloys the only environmental requirement is the presence of water in some form.

Stress Corrosion Cracking

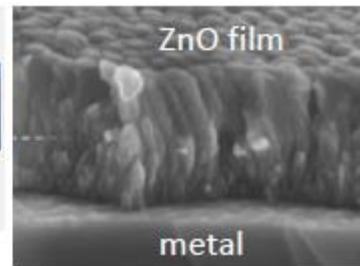
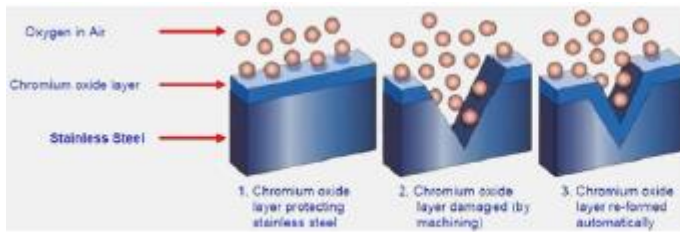


It is the growth of crack formation in a corrosive environment. It can lead to unexpected sudden failure of normally ductile metals subjected to a tensile stress, especially at high T.

← Scheme of stress corrosion crack propagation in a passive metal by quasi continuous, intermittent crack tip-slip activation followed by deformation-enhanced dissolution of metal at the crack tip. Note the role of a plastic deformation zone at the crack tip and protection of the crack walls by passivation, after crack growth has occurred.

The Passivating Oxide film

- Many metals display active corrosion behaviour when immersed in a corrosive solution, like Al, Ti, Cr and Zn.
- Other metals including many engineering alloys **carry a stable film of solid corrosion product**, frequently an **oxide**.
- If this film has **good cohesive strength and adhesion to the metal**, and **low ion conductivity**, the rate of metal dissolution is limited to the rate at which metal ions can pass through the film, often a factor of $10^3 - 10^6$ times less than the rate of corrosion of uncoated metals.
- This is the phenomenon of **passivity**, exhibited by **stainless steels (Cr)**, **nickel alloys**, **titanium (highly reactive)** and many similar materials and responsible for their utility as engineering materials in corrosive environments.



Uniform oxide film

If the metal film is dense and quite bonded with the substrate, the film is passivating, self-protecting the metal below.

Prevention of Corrosion

1. Weld rather than rivet (critical for galvanic and crevice corruptions);
2. Containers should be designed for easy draining;
3. Design for easy replacement of expected failures;
4. Avoid stresses in corrosion exposed components;
5. Avoid electrical contact between dissimilar materials;
6. Avoid heterogeneity (metals, vapor spaces, heat, stress);
7. Multiphase and particularly lamellar structure materials are prone to corrosion (exception is duplex S.S., two phased materials ferrite + austenite → used for corrosion resistance).

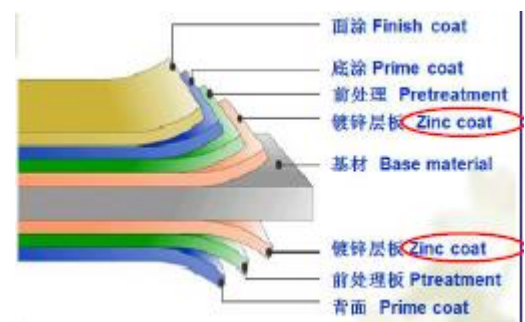
In order to avoid corrosion, we must have to consider: anode, cathode and electrolyte. The conductor is given by the metals themselves. If we have all three, we have corrosion. If one of these is missing, we will avoid corrosion. Another way is to combine two metals having similar electric potential → Corrosion will happen in any case, but we have to make the potential differences as small as possible.

Protection Methods

In the case of steels, coating the anode system with **Zn** or **Mg** will save the mechanical component from corrosion (ex. In galvanized sheet steels Zn or Mg sacrifice themselves to save Fe).

Coatings

- metallic – often dual function
- cladding
- flame spaying, vapor deposition
- electro-deposition (electroplating)
- hot-dipping (Zn hot dip coating)
- polymers



Ranks of reactivity of metals/alloys in seawater

More realistic and practical ranking.

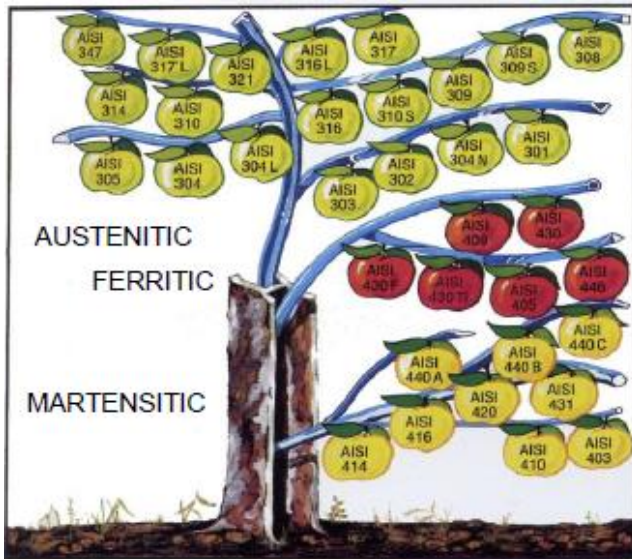
Alloys are the **top are cathodic** and unreactive, whereas those at the **bottom are most anodic**.

No voltages are provided.

Magnesium is the worst and Platinum is the best



Stainless Steels



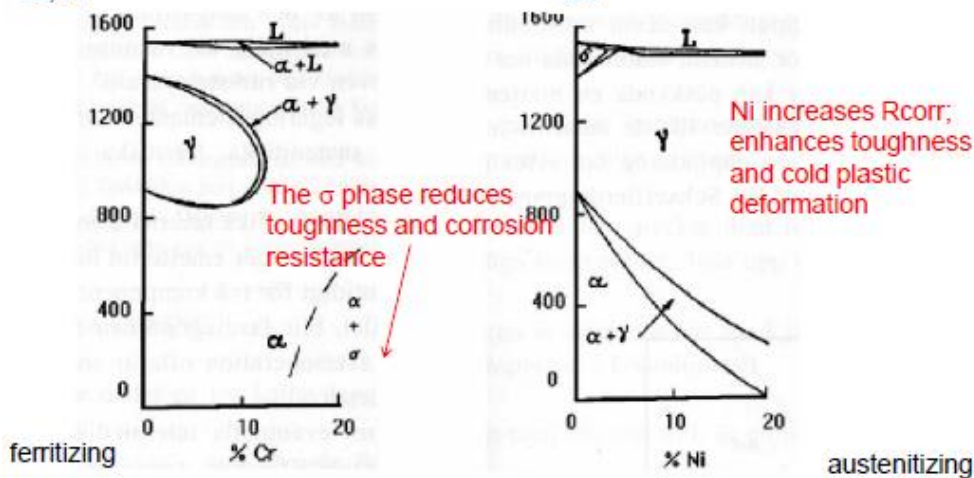
← big family of stainless steels.

Excellent corrosion resistance in stainless steel is due to high (at least 11%) **Chromium** forming chromium oxide on surface (to protect the material below). However, other key elements such as **Ni**, **Mo** and **N** may play a leading role in adjusting their properties required for specific applications.

Phase diagrams: Fe-Cr and Fe-Ni

Cr and Ni are the most important elements in austenitic stainless steels.

Cr limits the γ field (completely close); it has ferritizing action, such as Mo, Si, Nb, Ta, Al
Ni widens the γ field (stabilizes austenite) together with C, N, Mn, Cu



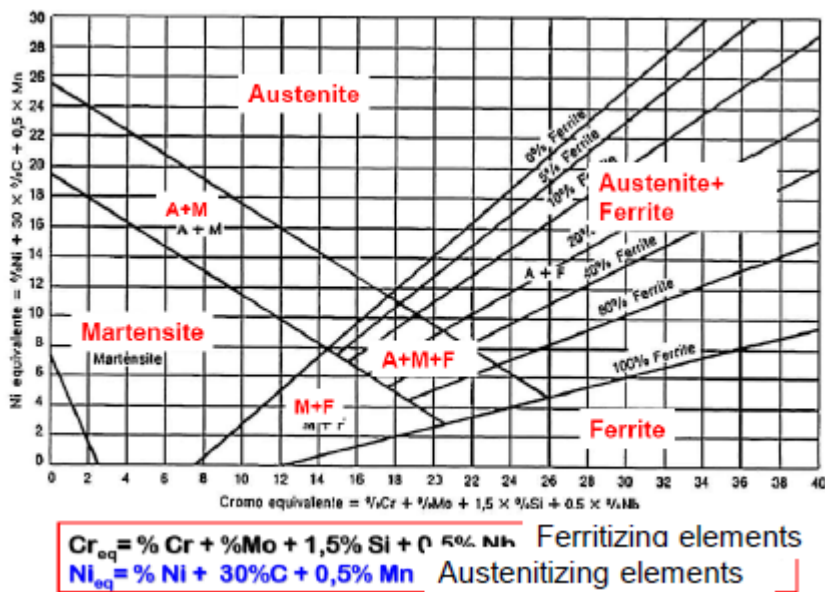
Classification of stainless steels

Alloy	C, wt %	Cr, wt %	Ni, wt %	Mo, wt %
FERRITIC	0,010 ÷ 0,12	10,5 ÷ 30	—	0,2 ÷ 2,5
AUSTENITIC	0,015 ÷ 0,15	16 ÷ 28	6 ÷ 32	2 ÷ 7
DUPLEX (austeno-ferritic)	0,030 ÷ 0,050	22 ÷ 28	3,5 ÷ 8	0,1 ÷ 4,5
MARTENSITIC	0,08 ÷ 1,2	11 ÷ 19	—	0,2 ÷ 1,5

We don't want austenite

Manufacturing of these alloys requires careful degassing of the liquid (use of Argon → AOD and VOD) since O may oxidize Cr which has high chemical affinity to O thereby losing their corrosion resistance potential.

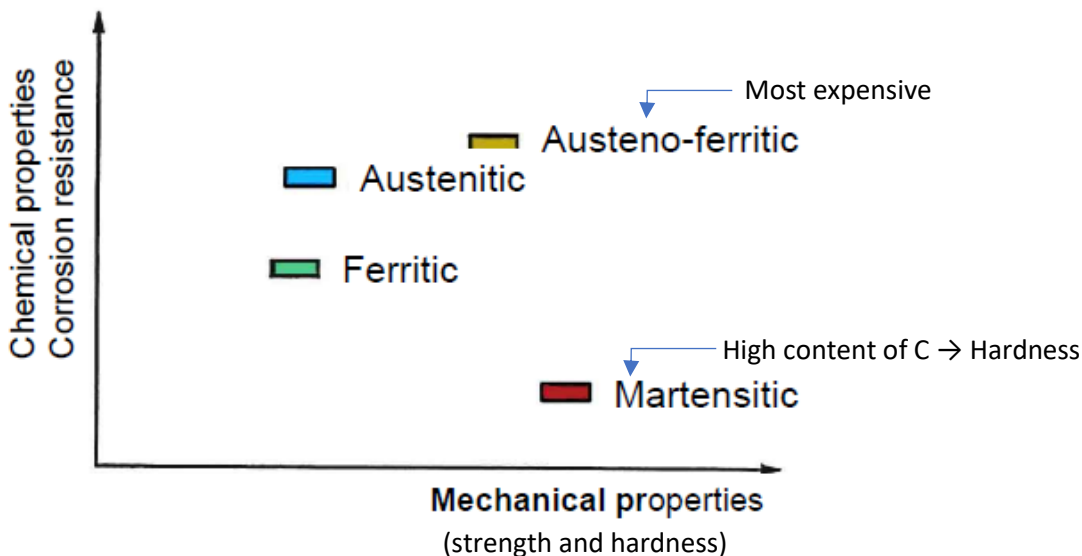
Schaeffler Diagram



← domain of all stainless steels.

Cr and Ni are the main elements but also other elements have important roles.

Chemical vs. mechanical properties of S.S.



If we want good corrosion resistance and less cost, we choose ferritic S.S.

Role of main alloy elements

Mo	↑ stability of passivating film especially in reducing environment, ↑ mechanical properties of austenitic especially at high T	Mo is a carburizing element so it forms very fine carbides. Because Mo is a refractory element and it has a high melting temperature.
Si	↑ Rox at high temperature	Rox → oxidation resistance. Si is also a refractory element.
Ti, Nb, Ta	carbide-forming elements, refine grains and Impede intergranular corrosion	
Al	As Si, ↑ Rox at high T in refractory steels	Al will oxidize at high temperature → it improves oxidation resistance.
Mn	In austenitic steels, may replace Ni as austenitizing element	Mn is the best austenitizing element.

Austenitic S.S. (Cr-Ni) – Series 3XX (in AISI standard)

The presence of Ni expands the austenitic range.

Advantages

- Excellent R_{corr}(corrosion resistance)
 - Excellent plastic deformation
 - High toughness
 - Absence of DBTT
 - Good R_{creep}
 - High R_{fatigue}
 - Strengthening by cold work
 - Good weldability
 - Paramagnetic properties
- Tensile strength: 559-759 MPa.

Limits

- High price (Ni and Cr are expensive)
- Low magnetic properties
- Suffer pitting corrosion

Represents 60%
of the overall production
of S.S.

Used for chemical equipment,
pressure vessels etc.

Nb prevents intergranular corrosion
if the alloy is to be used for welding.

Excellent plastic deformation → austenite is most ductile, best to be work hardened.

DBTT → Ductile to Brittle Transition Temperature.

Heat treatment is not possible because it changes the phase from austenitic to another one → different alloy.

The increase of strength can be achieved only by cold working and grain refinement.

Ferritic S.S. 12-30% Cr

Structure is mainly ferritic (BCC α): Cr extends α region and suppresses γ region.

Advantages

- Good R_{corr}, much higher than Martensitic SS
- higher R_{sc} than Austenitic SS
- Good plastic deformation (< than Austenitic SS)
- Ferromagnetic props
- < expensive than Austenitic SS
- High strength (517 MPa)

Limits

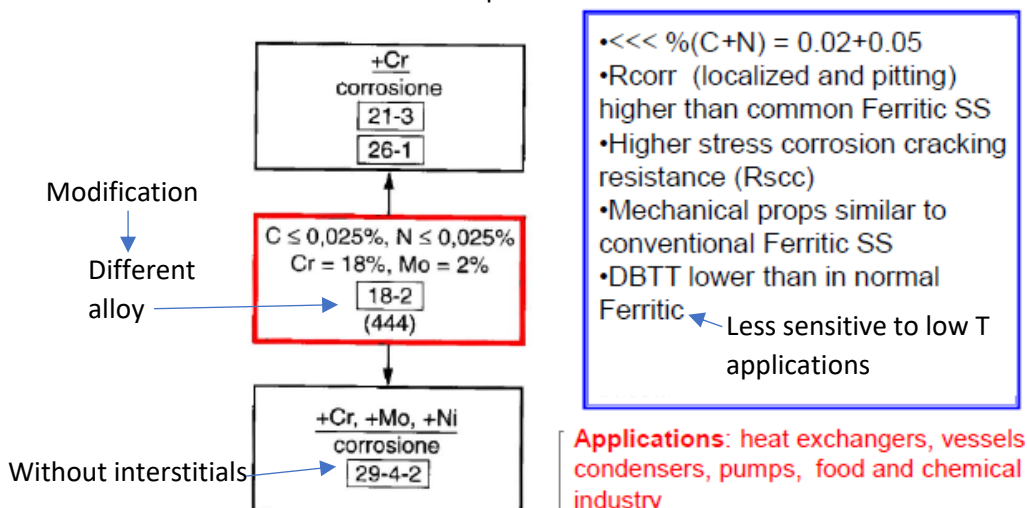
- Generally weldable but tendency to grain growth
- Presence of DBTT

R_{sc} → Resistance to Stress Corrosion Cracking → corrosion resistance while stressing the component.

It may become a little bit brittle at low T.

ELI – Ferritic S.S.

ELI stands for Extra Low Interstitials → improvements in ferritic S.S.



Martensitic S.S.

Martensite formed from **quenching** from austenite region (same as plain carbon steel); when C > 1%, **γ loop** is enlarged (C is an austenitizing element).

Advantages

Martensite looks more like ferrite

- High mechanical strength and hardness
- Ferromagnetic props

Limits

- < R_{corr} than others, espec. in chlorides, possible pitting corr. and stress corr. cracking

Martensite is brittle → crack propagation is faster with stress applied.

Heat treatment: Tensile strength: 517 MPa to 1966 MPa.

Applications: machine parts, pumps, steam turbine blades, valve parts, bolts and screws, cutters, surgery tools, extractive industry, bearings.

Stainless Steels: Weldability

Welding of FERRITIC SS

Possible formation of martensite in melting pool and HAZ.

Risk for grain growth (drop of resilience).

Carbides precipitation and possible formation of sigma phase.

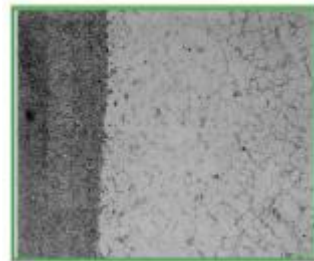
Welding of AUSTENITIC SS

Austenitic SS are most weldable among SS, as they have high initial toughness and good weldability because of the austenitic phase. However, they may experience precipitation of carbides in the melt pool or HAZ:

- Often precipitation occurs in HAZ because of high CR.

In contrast to other carbon steels, SS do not originate cracks after weld cooling.

Austenitic SS



Index of localized corrosion resistance

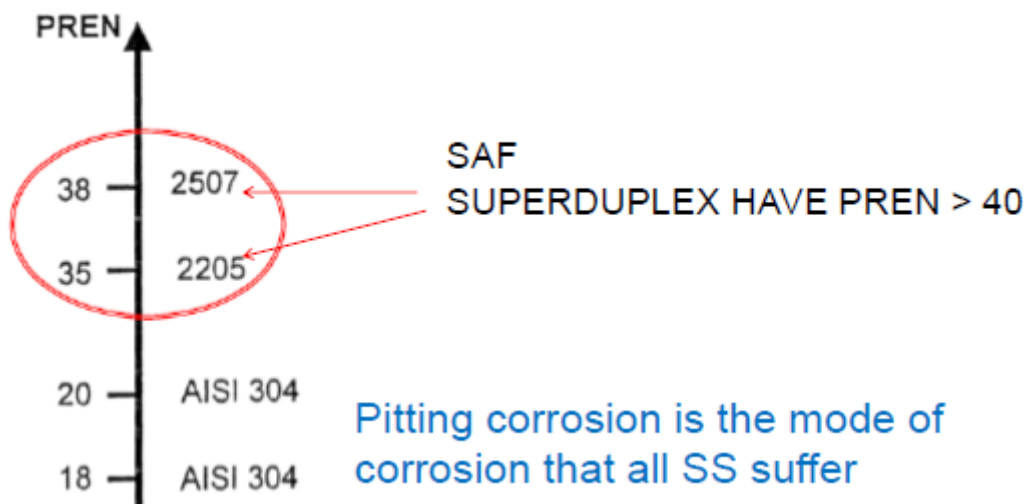
- Most of AUSTENITIC SS are susceptible of localized corrosion (e.g. crevice corrosion e pitting corrosion).
- To quantify the corrosion resistance of these alloys a corrosion resistance index has been defined (*Localized Corrosion Resistance Index - LCRI o Pitting Resistance Equivalent number – PREN*):

$$PREN = (A \times \%Cr) + (B \times \%Mo) + (C \times \%N) - (D \times \%Ni)$$

- Where A, B, C, and D are constants.
- According to Nadzhin-Wensley the **PREN** coefficients are:

$$PREN = \%Cr + 3,3\%(Mo+0,5\%W) + 16\%N$$

We have an increase of corrosion resistance thanks to the high content of Ni.



Typical S.S. Composition

		C	Cr	Ni	Mo	N	R (MPa)	A (%)
Austenitic	AISI 304	0.06	19	9	-	-	586	55
	AISI 302	0.15	18	9	-	-	620	50
	AISI 316	0.06	17	12	2.5	-	620	50
	AISI 304L	0.03	19	10	-	-	517	55
	AISI 316L	0.03	17	12	2.5	-	517	50
Ferritic	AISI 430	0.12	17	-	-	-	517	30
Martensitic	AISI 410	0.15	12	-	-	-	1310	15
Duplex	23-04	0.02	18	4	-	0.10	700	25
	22-05	0.02	22	5.5	3	0.14	800	25
	25-07	0.02	25	7	4	0.25	920	20
(Precipitation hardening)								
	17-4	0.07	17	4	0.4%Nb			10
	17-7	0.09	17	7	1%Al			6

AISI 304L and **AISI 316L** → they were the basic steels used in the past for corrosion resistance but they led to failure as pitting corrosion and intergranular crack formed during welding. The letter L indicates that there has been a modification of the original S.S. to reduce intergranular corrosion during welding. We can modify it also by introducing N as it increases corrosion resistance to pitting.

Precipitation hardening differs from others because the strengthening mechanism is by precipitation → highly controlled grain size and amount of precipitates can allow to have very high mechanical properties (they may even exceed 2 GPa).

Mechanical properties of S.S.

S.S.	σ_y (MPa)	%Elong
Ferritic	275-600	20-30
Austenitic	200-380	45-60
Duplex	315-565	20-40
Martensitic	275-1900	15-30
PH	1200-1800	5-15

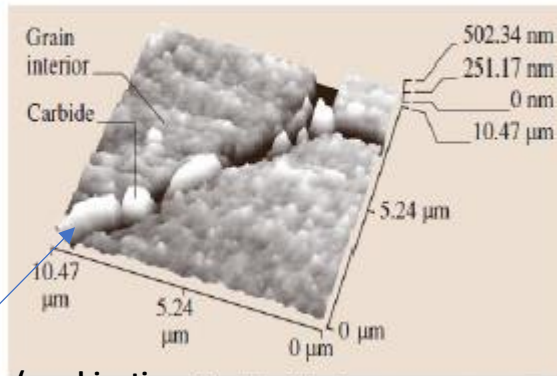
Summary of properties of S.S.

ALLOY GROUP	Magnetic Response (note 1)	Work Hardening Rate	Corrosion Resistance (note 2)	Hardenable	Ductility	High Temperature Resistance	Low Temperature Resistance (note 3)	Weldability
Austenitic	Generally No	Very High	High	By Cold Work	Very High	Very High	Very High	Very High
Duplex	Yes	Medium	Very High	No	Medium	Low	Medium	High
Ferritic	Yes	Medium	Medium	No	Medium	High	Low	Low
Martensitic	Yes	Medium	Medium	Quench & Temper	Low	Low	Low	Low
Precipitation Hardening	Yes	Medium	Medium	Age Hardening	Medium to Low	Low	Low	Low

Sensibilization of S.S. (in welding)

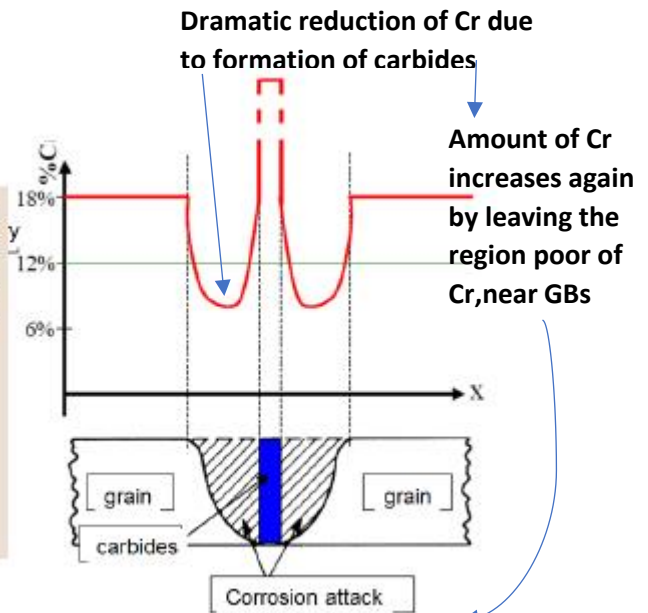
Sensibilization → Cr taken by C.

AFM photograph of a highly alloyed stainless steel undergoing intergranular corrosion, leaving behind chromium-rich carbides



Carbides (combination of small amount of C present in S.S and Cr) form at GBs → defects

Cr-depleted zone



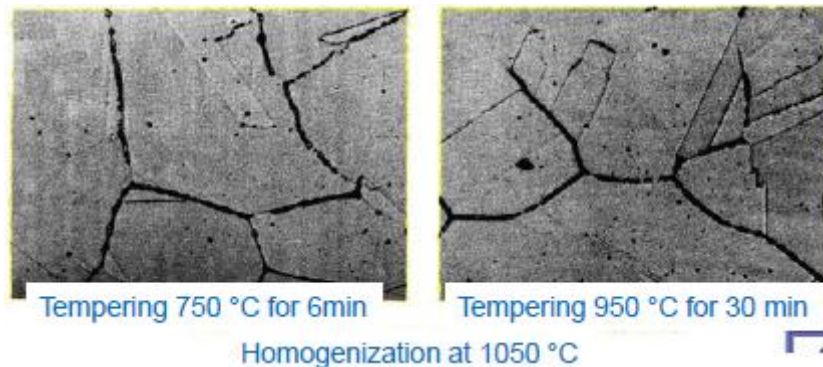
Corrosion will attack the region poor of Cr

Failure

Intergranular corrosion during welding of austenitic S.S.

In order to improve intergranular corrosion into austenitic S.S., we have to modify the original alloy in order to reduce to minimum the carbon content.

Sensibilization of austenitic SS (AISI 304)



Sensibilization of A.S.S. is generally induced by **welding, cutting, hot plastic deformation** (forging, relaxation heat treatment at high T):

It is caused by the **precipitation of chromium carbides** either along the slip bands after plastic deformation or grain boundaries in welding.

There are a few alloy systems in which very severe attack can occur at grain boundaries, usually as a result of **improper heat treatment**.

Austenitic stainless steels may become **sensitized** to intergranular attack if they are **heated in the range 500-800 °C**.

Chromium carbide precipitates at grain boundaries, and by removing large amounts of chromium is precipitated at grain boundaries, and by removing large amounts of chromium from solid solution renders the zone adjacent to the boundary particularly susceptible to attack.

This effect may be found in the **heat-affected zone adjacent to welds**, if too great a heat input has held the temperature of the zone in the sensitizing range for a sufficient period.

The problem is avoided either by **correct heat treatment** or more commonly by **lowering the C < 0.03 %** or stabilizing the carbon by the **addition of strong carbide-formers** such as Ti, Ta or Nb so that the carbon is no longer available for chromium carbide precipitation (AISI 316L, AISI 304L).

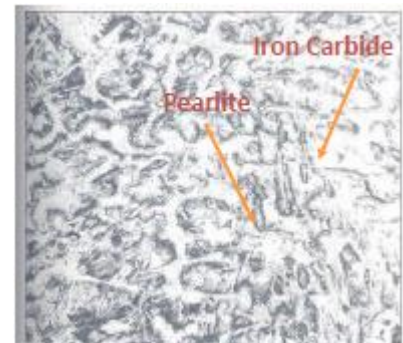
Cast Iron



- **General Properties:** 2-4.5% Carbon and 1-3% Si.
- **Ferrite** or **pearlite matrix** together with **graphite** as second phase (slow cooling)
- **Low melting point, ease of casting**, very fluid, **low shrinkage**, **easily machinable**.
- **Low impact resistance** and **ductility**.
- **Types of Cast Iron:**
 - White cast iron
 - Gray cast iron
 - Malleable cast iron
 - Ductile cast iron

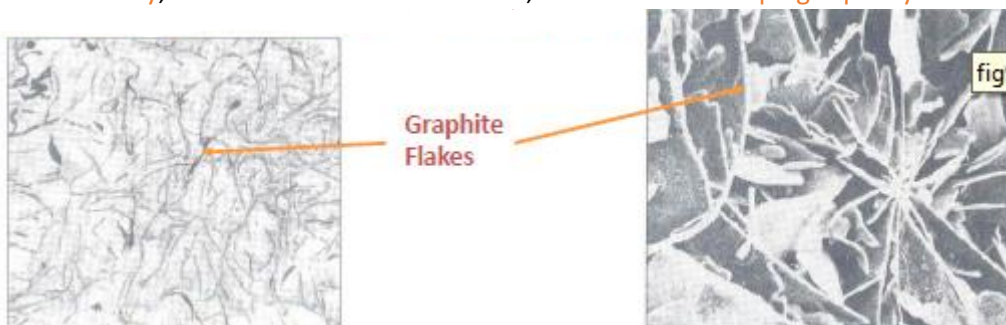
White Cast Iron

- Carbon is present in the form of **Iron Carbide** instead of **graphite** up on solidification.
- Fractured surface appears **white and crystalline**.
- Low C (2.5-3%) and Si (0.5-1.5%) content.
- Excellent wear resistance.
- High hardness.
- High brittleness.



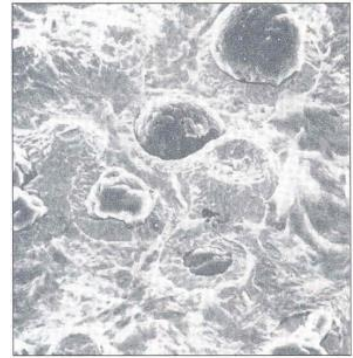
Gray Cast Iron

- C = 2.5-4% and Si = 1-3% (graphite former).
- C exceeds the amount that can be dissolved in austenite and precipitates as **graphite flakes**.
- Fractured surface appears gray
- Excellent **machinability**, **hardness** and **wear resistance**, and **vibration damping capacity**.



Ductile Cast Iron

- Has processing advantages of cast iron and **engineering advantages** of steel.
- Good fluidity, castability, machinability, and wear resistance.
- High strength, toughness, ductility and **hardenability** (due to **spherical nodules** of graphite).
- 3.4% C and 1.8-2.8% Si and low impurities.
- **Bull's eye** type microstructure.



Malleable Cast Iron

- 2-2.6% C and 1.1-1.6% Si.
- White cast iron is heated in **malleablizing furnace** to dislocate carbide into graphite.
- Irregular **nodules of graphite** are formed.
- Good castability, machinability, moderate strength, toughness and uniformity.

